

4-WEEK READMISSION PROJECT RESTART & PORTFOLIO PLAN

A self-guided daily plan to recover your MIMIC-IV project,
finish the modeling story, publish GitHub, and restart applications

Primary goal	Recover understanding first; do not rebuild from scratch.
Daily target	60–120 focused minutes. A 45-minute minimum version is acceptable.
Rule	Do not add new complexity until you can explain the existing decision.
End of Week 4	A project you can explain, a polished GitHub repo, updated CV/LinkedIn, and active applications.

Your project in one sentence

Build and evaluate a leakage-safe MIMIC-IV model for 30-day hospital readmission, comparing administrative, clinical, and physiologic/treatment information using interpretable and tree-based models.

Use this checklist every workday:

- ☐ Start by writing from memory for 5–10 minutes BEFORE opening old code.
- ☐ Then inspect the actual notebook/code and correct your notes.
- ☐ End by writing 3 things: What I did / Why I did it / What I would say in an interview.
- ☐ Mark questions with ??? instead of getting stuck. Resolve them during the designated review day.

WEEK 1 — RECOVER THE PROJECT STORY

Goal: understand the project again from cohort through modeling. No tuning this week.

Day 1 — Rebuild the big picture (60–90 min)

- ☐ Create a file called PROJECT_RECONSTRUCTION_NOTES.md or a paper notebook.
- ☐ Without opening code, write what you remember under: research question, population, unit of observation, target, features, split, models, metrics, results.
- ☐ Open your project folders and make a simple map: notebook/file → purpose → output.
- ☐ Write the end-to-end flow: raw MIMIC-IV → cohort → label → features → split → preprocessing → models → evaluation.
- ☐ Deliverable: one-page project map. Do not fix code today.

Day 2 — Cohort definition (75–120 min)

- ☐ Find the notebook/code that creates the analytic cohort.
- ☐ Write down exactly what one row represents: patient, admission, or something else.
- ☐ Record inclusion criteria and exclusion criteria exactly as implemented.
- ☐ Identify the key IDs used (for example subject_id and hadm_id) and what each means.
- ☐ Check row counts at major cohort steps and record them.
- ☐ Explain aloud: 'Who is in my study population, and why?'
- ☐ Deliverable: Cohort section of reconstruction notes + unresolved questions.

Day 3 — Reconstruct the 30-day readmission label (90–120 min)

- ☐ Find where next admission timing / days-to-next-admission / readmission label are created.
- ☐ Write the target definition in plain English and then as pseudocode.
- ☐ Check how same-day events, transfers, planned admissions, death, and the final observed admission are handled. Record only what your code actually does.
- ☐ Make a tiny 3–5 admission example on paper and manually determine the label.
- ☐ Identify variables used to CREATE the target that must later be excluded from predictors.
- ☐ Deliverable: exact target definition + label flow diagram.

Day 4 — Reconstruct Tier 1 features (60–90 min)

- ☐ List administrative/demographic features actually present in your final feature table.
- ☐ For every feature write: source table, meaning, prediction-time availability, missingness concern.
- ☐ Separate current-admission information from prior-utilization information.
- ☐ Check whether race/insurance/admission type were grouped and find the mapping logic.
- ☐ Deliverable: Tier 1 feature dictionary.

Day 5 — Reconstruct Tier 2 features (75–120 min)

- ☐ List clinical burden and care-intensity variables actually implemented.
- ☐ Review LOS, ICU flag, diagnosis/comorbidity variables, procedures, and discharge context if present.
- ☐ Distinguish 'planned feature' from 'implemented feature'—do not assume your earlier plan was completed.
- ☐ Check whether Charlson/comorbidity measures were actually created; mark unfinished items clearly.
- ☐ Deliverable: Tier 2 feature dictionary + missing/unfinished feature list.

Day 6 — Reconstruct Tier 3 features (90–120 min)

- ☐ Review labs, vitals, medications, polypharmacy, and missingness indicators actually implemented.
- ☐ For each lab/vital, record the aggregation window and summary statistic (first/last/min/max/mean/etc.) if used.
- ☐ Revisit the SPO2/missingness issue: write why missingness may encode care setting rather than random absence.
- ☐ Check for any feature whose timing could accidentally include information after your intended prediction point.
- ☐ Deliverable: Tier 3 feature dictionary + timing/missingness notes.

Day 7 — Weekly recall test (45–75 min)

- ☐ Close all notebooks. On blank paper explain the project from research question through features.
- ☐ Draw the pipeline from memory.

- ☐ Answer: What is the cohort? What is the target? What is one row? What are Tier 1/2/3? Where can leakage occur?
- ☐ Open your notes only after answering and correct yourself.
- ☐ Create a 'Top 10 things I had forgotten' list.
- ☐ STOP condition: if you cannot explain cohort + target confidently, repeat Days 2–3 before moving on.

WEEK 2 — RECOVER THE ML & RESULTS

Goal: understand leakage prevention, splitting, preprocessing, models, metrics, and existing results.

Day 8 — Leakage audit (75–120 min)

- ☐ Find the final modeling feature list and every dropped column.
- ☐ For each leakage-related drop, write WHY it is leakage or inappropriate at prediction time.
- ☐ Pay special attention to next_admittime, days_to_next_admit, mortality/death fields, ED timing/provider fields, and raw variables replaced by grouped versions.
- ☐ Create two columns: Safe at prediction time? / Reason.
- ☐ Deliverable: leakage audit table suitable for future README/interviews.

Day 9 — Train/validation/test split (60–90 min)

- ☐ Find the exact split code and random seed.
- ☐ Confirm whether subject_id grouping prevents the same patient from crossing datasets.
- ☐ Record train/validation/test sizes and positive readmission rate in each split.
- ☐ Explain why a row-level random split could produce optimistic results.
- ☐ Deliverable: data-split summary with counts and target prevalence.

Day 10 — Preprocessing pipeline (75–120 min)

- ☐ Identify numeric, categorical, and binary feature groups.
- ☐ Trace imputation, one-hot encoding, scaling, and any other transformers.
- ☐ Write which transformations apply to which models.
- ☐ Confirm transformations are fit on training data through a pipeline rather than on the full dataset.
- ☐ Explain ColumnTransformer/Pipeline in plain English without code.
- ☐ Deliverable: preprocessing diagram + feature-type list.

Day 11 — Logistic Regression (60–90 min)

- ☐ Rerun or inspect the LR baseline only.
- ☐ Record parameters, ROC-AUC, PR-AUC, and any other metrics already produced.
- ☐ Review why scaling matters for LR.
- ☐ Identify the role of LR: interpretable baseline, not necessarily the winning model.
- ☐ Deliverable: one LR model card: inputs / preprocessing / metrics / interpretation.

Day 12 — Random Forest & XGBoost (90–120 min)

- ☐ Review RF and XGBoost separately; record the important parameters actually used.
- ☐ Record validation/test metrics and compare them with LR.
- ☐ Write 2–3 reasons tree models may capture relationships LR misses.
- ☐ Do not tune yet. First determine whether the existing models reproduce successfully.
- ☐ Deliverable: RF and XGB model cards + comparison table.

Day 13 — Evaluation, SHAP, calibration & thresholds (90–120 min)

- ☐ Locate ROC curves, confusion matrices, SHAP plots, threshold analysis, and any calibration work.
- ☐ Record what exists versus what was only planned.
- ☐ For SHAP, identify the top features and write what the model appears to be learning—without claiming causality.
- ☐ Check whether PR-AUC is reported; readmission may be imbalanced, so do not rely only on accuracy.
- ☐ Add calibration/threshold analysis to the unfinished list if not completed.
- ☐ Deliverable: results inventory + 3–5 defensible findings.

Day 14 — Mock interview #1 (60–90 min)

- ☐ Give yourself a 5-minute spoken project presentation with no notes.
- ☐ Answer: Why readmission? Why MIMIC-IV? How did you prevent leakage? Why patient-level splitting? Why LR/RF/XGB? Why ROC-AUC and PR-AUC? What did SHAP tell you?
- ☐ Record questions you cannot answer.

- ☐ Create a one-page 'Interview Cheat Sheet'.
- ☐ Decision point: identify only the modeling work that is genuinely needed before publication.

WEEK 3 — CLEAN, COMPLETE & PUBLISH GITHUB

Goal: turn existing work into a reproducible portfolio project. Do targeted modeling only.

Day 15 — Decide the minimum final analysis (60–90 min)

- ☐ Create MUST / NICE-TO-HAVE / NOT NOW columns.
- ☐ MUST: reproducible split, baseline models, core metrics, clear leakage logic, one interpretation method, limitations.
- ☐ Consider calibration and threshold analysis as higher-value additions than exhaustive hyperparameter search.
- ☐ Decide whether Tier 1 vs Tier 1+2 vs Tier 1+2+3 comparison is feasible with your existing feature table.
- ☐ Set a hard scope for the remaining two weeks.

Day 16 — Targeted model tuning (90–120 min)

- ☐ Only if baseline pipeline is stable: choose ONE main model (likely XGBoost or the strongest existing candidate) for modest tuning.
- ☐ Select a small, justified parameter search rather than a huge grid.
- ☐ Use validation/CV appropriately; keep the test set untouched until final evaluation.
- ☐ Record baseline vs tuned performance and runtime.
- ☐ If tuning adds negligible value, keep the simpler configuration and say so.
- ☐ Deliverable: tuning experiment table.

Day 17 — Final evaluation (90–120 min)

- ☐ Freeze the chosen pipeline/model.
- ☐ Evaluate once on the held-out test set.
- ☐ Create final ROC-AUC and PR-AUC comparison table.
- ☐ Add calibration plot/metric if feasible and clinically relevant threshold examples.
- ☐ Save clean figures with readable titles/labels.
- ☐ Deliverable: final_results table + publication-ready figures.

Day 18 — Interpretation & limitations (75–120 min)

- ☐ Produce/review SHAP summary for the final tree model; for LR consider coefficient interpretation where appropriate.
- ☐ Write 5 findings in cautious language: association/prediction, not causation.
- ☐ Write limitations: single-center EHR, retrospective data, label definition, missingness, temporal/generalizability concerns, no prospective clinical validation.
- ☐ Write what you would do next with more time.
- ☐ Deliverable: Results + Limitations draft.

Day 19 — GitHub cleanup (90–120 min)

- ☐ Create a clean repository structure: README, notebooks, src if applicable, figures, requirements/environment notes.
- ☐ Remove temporary cells, dead code, giant outputs, local absolute paths, and secrets.
- ☐ Do NOT upload MIMIC-IV patient-level/source data.
- ☐ Add a data-access note explaining that users must obtain MIMIC-IV through its authorized source.
- ☐ Check notebooks run in a logical order.

Day 20 — Write the README (90–120 min)

- ☐ Write sections: Overview, Clinical Question, Data, Cohort, Outcome, Features, Leakage Prevention, Modeling, Evaluation, Results, Interpretation, Limitations, Reproducibility.
- ☐ Put the best model comparison table and 1–3 useful figures near the top/middle.
- ☐ Explain the project for a hiring manager first; technical detail comes second.
- ☐ Include your three-tier feature framework only to the extent actually implemented.
- ☐ Deliverable: complete README draft.

Day 21 — GitHub recruiter test (60–90 min)

- ☐ Open the repo as if you know nothing about it. Give yourself 2 minutes.
- ☐ Can you identify the problem, data, methods, result, and your contribution immediately?

- ☐ Check every README link and figure.
- ☐ Ask: 'Could I explain every claim shown here in an interview?' Remove or qualify anything you cannot defend.
- ☐ Deliverable: GitHub v1 ready to share.

WEEK 4 — CV, LINKEDIN, INTERVIEW & APPLICATIONS

Goal: package the project professionally and begin applying before everything feels perfect.

Day 22 — Write project résumé bullets (60–90 min)

- ☐ Draft 3–4 bullets using action + method + result + significance.
- ☐ Include MIMIC-IV, 30-day readmission, leakage-safe patient-level splitting, Python/sklearn/XGBoost, and actual metrics only after verifying them.
- ☐ Include SHAP/calibration/threshold work only if actually completed.
- ☐ Keep a longer master version, then shorten for individual jobs.
- ☐ Deliverable: final project section for master CV.

Day 23 — Update the full CV (90–120 min)

- ☐ Integrate the readmission project with your prior research/analytics experience.
- ☐ Move the most job-relevant skills/projects high enough to be noticed quickly.
- ☐ Verify every technical skill listed is something you can discuss.
- ☐ Create one healthcare-data version as your base CV.
- ☐ Deliverable: application-ready CV v1.

Day 24 — LinkedIn update (60–90 min)

- ☐ Update headline toward healthcare data / analytics / data science.
- ☐ Rewrite About section around your transition story, quantitative/research background, and current healthcare ML work.
- ☐ Add the MIMIC-IV project under Projects/Featured and link GitHub.
- ☐ Align Skills with what appears in your CV: Python, pandas, sklearn, SQL as appropriate, XGBoost, SHAP, healthcare analytics.
- ☐ Deliverable: LinkedIn profile aligned with CV/GitHub.

Day 25 — Build application system (60–90 min)

- ☐ Create a simple tracker: company, role, link, date, fit, CV version, contact, status, follow-up date.
- ☐ Define 2–3 target role families rather than applying randomly.
- ☐ Create a base cover-letter story but do not over-customize every application.
- ☐ Set a sustainable weekly application target for after this four-week sprint.
- ☐ Deliverable: job tracker + target-role criteria.

Day 26 — Apply to first roles (90–120 min)

- ☐ Select 2–3 roles with reasonable fit; do not wait for perfect fit.
- ☐ Tailor CV keywords truthfully to each posting.
- ☐ Submit applications and record them immediately.
- ☐ Save job descriptions because postings may disappear before interviews.
- ☐ Deliverable: first applications sent.

Day 27 — Mock interview #2 + coding refresh (90–120 min)

- ☐ Give a 10-minute project presentation: problem → cohort → target → leakage → features → models → results → limitations.
- ☐ Practice explaining one code pipeline and one SHAP/result figure.
- ☐ Do a short Python/pandas/SQL refresh focused on operations you actually used in the project.
- ☐ Write concise answers to the 10 most likely project questions.
- ☐ Deliverable: interview Q&A sheet.

Day 28 — Final review & next 30-day rhythm (60–90 min)

- ☐ Audit the four-week outputs: reconstruction notes, model results, GitHub, CV, LinkedIn, applications.
- ☐ List remaining technical debt separately so it does not block applications.
- ☐ Choose a continuing weekly rhythm: applications + interview practice + one project improvement block.
- ☐ Give your project presentation one final time without notes.

- ☐ Celebrate completion by writing: 'What I can now explain that I could not explain four weeks ago.'
- ☐ Deliverable: next-month plan and a finished restart phase.

WEEKLY SCORECARD

Week	Recall	Technical	Portfolio	Career	Done?
1	Can explain cohort/target/features	Project map + dictionaries	—	—	☐
2	Can explain ML choices/results	Leakage/split/model cards	—	Interview #1	☐
3	Can defend final analysis	Final results/tuning	GitHub v1	—	☐
4	10-min project story	Coding refresh	GitHub final	CV + LinkedIn + applications	☐

DO NOT LET THESE DELAY APPLICATIONS

- ☐ Building a sophisticated personal website. GitHub + LinkedIn are enough initially.
- ☐ Trying every ML algorithm.
- ☐ Large hyperparameter searches for tiny metric gains.
- ☐ Perfecting every notebook before sending any CV.
- ☐ Learning new tools unrelated to the jobs you are applying for.
- ☐ Making causal claims from predictive feature importance.

YOUR END-OF-MONTH PORTFOLIO CHECKLIST

- ☐ I can explain my cohort and exact 30-day readmission definition.
- ☐ I can explain the leakage risks and why I used patient-level splitting.
- ☐ I can explain preprocessing and why it belongs inside the modeling pipeline.
- ☐ I can compare Logistic Regression, Random Forest, and XGBoost.
- ☐ I know my verified ROC-AUC / PR-AUC results.
- ☐ I can explain my SHAP results without confusing prediction with causation.
- ☐ I can discuss calibration/threshold choices or clearly state that they are future work.
- ☐ My GitHub README tells the complete project story.
- ☐ No restricted MIMIC-IV patient-level data are published.
- ☐ My CV and LinkedIn link to the project.
- ☐ I have started applying.
- ☐ I can give a 5–10 minute project explanation without opening the notebook.

Principle for the month: RECALL → VERIFY → EXPLAIN → PUBLISH → APPLY